

Problem Sheet: Chain Rule and Substitutions 2

Solution

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Exercise 1. Differentiate the following functions.

1. $\sin(\cos(3x))$
2. $e^{2\sin(4x)}$
3. $\cos(\ln(5x))$
4. $(\ln(2x))^3$

Solution. 1. Differentiating $\sin(\cos(3x))$

Let

$$y = \sin(\cos(3x)), \quad u = \cos(3x)$$

so that $y = \sin u$.

Then

$$\begin{aligned}\frac{dy}{du} &= \cos u \\ \frac{du}{dx} &= -3\sin(3x)\end{aligned}$$

Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \cos u \cdot (-3\sin(3x)) \\ &= -3\sin(3x) \cos(\cos(3x))\end{aligned}$$

2. Differentiating $e^{2\sin(4x)}$

Let

$$y = e^{2\sin(4x)}, \quad u = 2\sin(4x)$$

so that $y = e^u$.

Then

$$\begin{aligned}\frac{dy}{du} &= e^u \\ \frac{du}{dx} &= 8\cos(4x)\end{aligned}$$

Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= e^u \cdot 8\cos(4x) \\ &= 8e^{2\sin(4x)} \cos(4x)\end{aligned}$$

3. Differentiating $\cos(\ln(5x))$

Let

$$y = \cos(\ln(5x)), \quad u = \ln(5x)$$

so that $y = \cos u$.

Then

$$\begin{aligned} \frac{dy}{du} &= -\sin u \\ \frac{du}{dx} &= \frac{1}{x} \end{aligned}$$

Therefore

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= -\sin u \cdot \frac{1}{x} \\ &= -\frac{\sin(\ln(5x))}{x} \end{aligned}$$

4. Differentiating $(\ln(2x))^3$

Let

$$y = (\ln(2x))^3, \quad u = \ln(2x)$$

so that $y = u^3$.

Then

$$\begin{aligned} \frac{dy}{du} &= 3u^2 \\ \frac{du}{dx} &= \frac{1}{x} \end{aligned}$$

Therefore

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= 3u^2 \cdot \frac{1}{x} \\ &= \frac{3(\ln(2x))^2}{x} \end{aligned}$$

□

Exercise 2. Let $f(x) = e^{3 \sin x} + \sin(x^2)$. Find $f'(x)$ and $f''(x)$.

Solution. Let

$$f(x) = e^{3 \sin x}, \quad g(x) = \sin(x^2).$$

Then

$$(e^{3 \sin x} + \sin(x^2))' = f'(x) + g'(x).$$

For $f(x)$, let $y = e^{3 \sin x}$ and $u = 3 \sin x$. Then $y = e^u$, so

$$\begin{aligned} \frac{dy}{du} &= e^u \\ \frac{du}{dx} &= 3 \cos x \end{aligned}$$

and therefore

$$f'(x) = 3e^{3 \sin x} \cos x.$$

For $g(x)$, let $y = \sin(x^2)$ and $u = x^2$. Then $y = \sin u$, so

$$\begin{aligned}\frac{dy}{du} &= \cos u \\ \frac{du}{dx} &= 2x\end{aligned}$$

and therefore

$$g'(x) = 2x \cos(x^2).$$

Hence the first derivative is

$$(e^{3 \sin x} + \sin(x^2))' = 3e^{3 \sin x} \cos x + 2x \cos(x^2).$$

Now differentiate again.

For $3e^{3 \sin x} \cos x$, use the product rule. Let

$$u = 3e^{3 \sin x}, \quad v = \cos x.$$

Then

$$v' = -\sin x$$

and to find u' we again use the chain rule:

$$u' = 3(e^{3 \sin x})' = 3(3e^{3 \sin x} \cos x) = 9e^{3 \sin x} \cos x.$$

Therefore

$$\begin{aligned}(uv)' &= uv' + u'v \\ &= 3e^{3 \sin x}(-\sin x) + 9e^{3 \sin x} \cos x \cdot \cos x \\ &= -3e^{3 \sin x} \sin x + 9e^{3 \sin x} \cos^2 x\end{aligned}$$

For $2x \cos(x^2)$, use the product rule again. Let

$$u = 2x, \quad v = \cos(x^2).$$

Then

$$u' = 2$$

and for v' let $w = x^2$, so that $v = \cos w$. Then

$$\begin{aligned}\frac{dv}{dw} &= -\sin w \\ \frac{dw}{dx} &= 2x\end{aligned}$$

so

$$v' = -2x \sin(x^2).$$

Therefore

$$\begin{aligned}(uv)' &= uv' + u'v \\ &= 2x(-2x \sin(x^2)) + 2 \cos(x^2) \\ &= -4x^2 \sin(x^2) + 2 \cos(x^2)\end{aligned}$$

Hence the second derivative is

$$\begin{aligned}(3e^{3 \sin x} \cos x + 2x \cos(x^2))' &= -3e^{3 \sin x} \sin x + 9e^{3 \sin x} \cos^2 x \\ &\quad - 4x^2 \sin(x^2) + 2 \cos(x^2)\end{aligned}$$

□

Exercise 3. For each integral below, state a suitable substitution before evaluating the indefinite integral.

1. $\int x e^{x^2} dx$

2. $\int \frac{4x}{1+2x^2} dx$

3. $\int 6x \sqrt{1-3x^2} dx$

Solution. 1. Integrating $\int x e^{x^2} dx$

We use the substitution

$$u = x^2.$$

Then

$$du = 2x dx$$

so

$$x dx = \frac{1}{2} du.$$

Therefore

$$\begin{aligned} \int x e^{x^2} dx &= \frac{1}{2} \int e^u du \\ &= \frac{1}{2} e^u + c \\ &= \frac{1}{2} e^{x^2} + c. \end{aligned}$$

2. Integrating $\int \frac{4x}{1+2x^2} dx$

We use the substitution

$$u = 1 + 2x^2.$$

Then

$$du = 4x dx.$$

Therefore

$$\begin{aligned} \int \frac{4x}{1+2x^2} dx &= \int \frac{1}{u} du \\ &= \ln |u| + c \\ &= \ln |1 + 2x^2| + c. \end{aligned}$$

3. Integrating $\int 6x \sqrt{1-3x^2} dx$

We use the substitution

$$u = 1 - 3x^2.$$

Then

$$du = -6x dx,$$

so

$$6x dx = -du.$$

Therefore

$$\begin{aligned}\int 6x\sqrt{1-3x^2} \, dx &= -\int \sqrt{u} \, du \\ &= -\int u^{1/2} \, du \\ &= -\frac{2}{3}u^{3/2} + c \\ &= -\frac{2}{3}(1-3x^2)^{3/2} + c.\end{aligned}$$

□